Word-count declaration

Manuscript: *A Comparability Protocol for Benchmarking Relational Database Access Layers in Express.js* (submission build `ist/ist_main.tex`).

Counted with `texcount` on the seven body sections plus the structured abstract. The IST *Guide for Authors* sets the limit for a research paper at **15,000 words**, and states that **"references and appendices are part of the submission and count against the total number of words, and figures and tables count 200 words each."** This count applies that rule exactly: the reference list is included and each of the eight main-text floats counts 200 words. The paper has **no appendices** — all supplementary material is in the separate `supplement.pdf`, which under Elsevier's policy is online supplementary material, not an appendix, and is not part of the manuscript word count. The mandatory declaration sections (CRediT, Declaration of competing interest, Funding, Data availability, and the generative-AI declaration, ~300 words) are required metadata, not article content or appendices, and are excluded per Elsevier convention.

Component	Count
Body text (7 sections)	10649
Structured abstract	299
Tables and figures in the main text (8 × 200)	1,600
Reference list (62 entries)	1,848
Total (IST rule)	**14396**

This is under the journal's 15,000-word limit (with the abstract counted; excluding the abstract it is 14097). The structured abstract is 299 words (under the journal's 300-word structured-abstract limit). Thirty-two tables and two figures are placed in the numbered online supplement (`supplement.pdf`, Supplement Tables S1–S32 and Supplement Figures S1–S2), which is submitted with the manuscript and archived under the same Zenodo DOI as the code and data; supplement floats are not part of the main-text count. A later revision added per-cell bootstrap 95% CIs to the p99 column of every pattern table and a paired p99 significance table (Supplement Table S30), delivering the p99 inference the outcomes table lists as primary; these widen existing tables and add one supplement float, not a main-text float.

In this revision (round 5) the entire benchmark was re-run on the latest stable library versions (Prisma 7, Express 5, etc.); the manuscript was repositioned as a reproducible benchmark and a configuration-specific comparison; the body prose was cut by roughly 1,800 words to remove repeated conclusions and filler (the five recurring findings are now stated once, in the results section, and cross-referenced elsewhere); the secondary access patterns (point read, keyset range scan) were reduced to lean summaries with their full tables in the supplement; the controlled/varied/uncontrolled factor table was moved to the supplement (Table S27); and five peripheral citations (general NoSQL and OLTP benchmark context) were dropped, taking the reference list from 70 to 62 entries. Six single-host experiments added earlier in the round (utilization-controlled open-loop, multi-worker cluster, per-layer pool frontier on both engines, mixed read/write, replicated fan-out, and a longer-run sensitivity check) remain as Supplement Tables S21–S26 with a one-sentence pointer each in the body.

Two later patch revisions left the reference list at 62 entries: the re-review revision (v1.5.1) dropped one peripheral citation, and v1.5.2 added one methodological citation (Papadopoulos et al. 2021, *IEEE Transactions on Software Engineering*, on reproducible performance-evaluation principles) anchoring the paper's perishability thesis. That citation adds no body prose (a bare `\{}cite`), so the only change to the count is the single reference entry (reference list 1,842 → 1,848 words, total 14,917 → 14,923).

Revision round 6 retracted the causal same-SQL attribution (now a bounding contrast), renamed the "idiomatic" estimand, separated the capacity/overload/matched-utilization quantities, tempered the perishability claim, and added the required statistical disclosures (Kendall's τ, an ex-MikroORM robustness note, a layer×engine interaction table, and an insert-dispersion figure, all in the supplement). The condensed Prisma case study and the removal of repeated restatements offset most of these additions, so the body changed marginally (11,376 to 11,388) and the abstract was tightened from 313 to 298 words; the total is 14,934. The longer-window tail sensitivity (Table S23), interaction magnitude (Table S28), and insert dispersion (Figure S1) are the new supplement floats.

A subsequent round-6 clarification renamed the primary 50-connection matrix from "saturated" to a "fixed 50-connection high-load operating point," since the throughput knee is demonstrated (Supplement Figure S2) only for the deep fetch, while the ten-connection pool—not a per-pattern knee—is the binding high-load constraint (Section 3, Controls); the per-layer/pool/open-loop "saturat*" uses are unchanged. The rename plus its one-clause justification added a net 15 body words (offset by tightening three tail-latency sentences) and one abstract word, taking the body to 11,444 and the total to 14,991.

A further round-6 clarification softened the causal claim that each layer is "single-core-bound" / "multi-core-bound": the equal-CPU and multi-worker controls show that extra application cores do not raise a process's throughput (so the ranking is not a core-budget artifact), but do not prove single-core CPU is the causal bottleneck. The label was replaced by that observation at every prose site and in the equal-CPU and cluster table captions (the latter also corrected: its own data shows Prisma flat across 1/2/4 cores, so "Prisma requires roughly four cores to approach pg" was wrong—the four are cluster workers, not cores). Merging a now-redundant sentence offset the reframe, so the body is 11,443 and the total 14,990; the CPU-utilization measurements ("$\{}approx$110\{}% of one core") are unchanged.

A final round-6 clarification separated the *fact* that an earlier Prisma result did not reproduce on the re-run from the *hypothesis* that Prisma's version/architecture change caused it: the Discussion now states the non-reproduction as a fact and its cause as an explicitly hedged hypothesis (the vanished premium was Prisma-specific and only Prisma changed architecture, so its Rust-free rewrite is the likeliest driver, but the re-freeze advanced the whole toolchain at once, so it cannot be isolated), and the intro/conclusion headline sentences now say the result was retired "on newer versions" rather than attributing it to "Prisma's Rust-free successor." Trimming the paragraph's lead-in and a parenthetical offset most of the hedge, taking the body to 11,447 and the total to 14,994.

A final round-6 terminology change replaced the over-claiming label "idiomatic" (~53 occurrences) with "documentation-selected" throughout, with lighter synonyms ("documented default", "default") where it read clunkily. The methodology already disclaimed the connotation ("Documentation order is a reproducible selection heuristic, not evidence that the chosen API is performance-optimal or the most common production choice"); this makes the terminology consistent with that definition. The "default-configuration" estimand name is unchanged; the generated table captions (sameplan, altloading, insert-dispersion) and the replication-package `METHODOLOGY.md` were updated to match. Dropping the now-redundant "used idiomatically" disclaimer clause took the body to 11,437 and the total to 14,984.

A final round-6 change simplified the statistical emphasis: the results subsection "Measurement stability and significance" was renamed "...and effect sizes" and rewritten to lead with geometric-mean ratios, bootstrap CIs, and practical significance (the narrowest adjacent step, 1.05, sits at the ±5% equivalence margin — statistically distinguishable yet practically marginal; the up-to-7× spread is decisive), with the post-hoc adjacent-pair permutation/Wilcoxon/Bonferroni tests compressed to a secondary, descriptive robustness note. The conclusion, methodology, and threats sentences were aligned to the same hierarchy (effect sizes and intervals primary; adjacent-rank tests secondary and descriptive). No statistic was added or removed; the significance table (Table 7) stays in the main text. Compressing the p-value enumeration offset the added practical-significance prose, so the body is 11,435 and the total 14,982.

A follow-on round-6 change reported the layer×engine interaction by its magnitude rather than its p-value: the RQ2 paragraph now leads with the per-layer PostgreSQL÷MySQL throughput ratios (reads a narrow $\{}approx$1.0–1.6× band, the insert scattering 1.6–3.2× and reordering layers across engines), with the blocked permutation test demoted to confirming the interaction is nonzero (its floor p says nothing about size); the outcomes-table row foregrounds the effect ratios (Table S28). The added magnitude detail took the body to 11,444 and the total to 14,991.

A final round-6 change softened the novelty claims to match the (explicitly non-systematic) scoping search: the one bald claim, "the combination missing from prior art" (introduction), became "a combination not identified in our documented search," and "a design choice absent from every access-layer benchmark surveyed above" (related work) became "not found in the access-layer benchmarks surveyed above." The four-axis gap statement and the related-work positioning were already scoped ("In the sources we searched…"; "a structured scoping search (not a systematic review)"). Net +3 words; the body is 11,447 and the total 14,994.

A final round-6 change stated explicit library inclusion/exclusion criteria in the methodology (each layer is maintained, in common use, occupies a distinct taxonomy tier, and — for the portable tiers — runs against both engines), with the reasons the named candidates were omitted: PostgreSQL-only clients (Slonik, pg-promise) are non-portable, and Kysely is a query builder already represented by Knex. The full criteria and a per-candidate table are in the replication package (`experiments/METHODOLOGY.md`, not word-counted). The addition was offset by removing a redundant tier-classification clause (Drizzle's placement, already given in the layer list) and tightening the vendor-independence sentence, so the net change is +1 word: the body is 11,448 and the total 14,995.

A final round-6 change added a short **artifact-reproducibility table** (Supplement Table S31: version/commit and DOI, software and hardware requirements, the run commands, the time and resources, and which results are regenerated automatically from the archived raw data), with a one-line pointer in the Data Availability section. Both are outside the counted body — the table is a supplement float (uncounted) and Data Availability is back-matter, not one of the seven counted sections — so the declared total is unchanged at 14,995. That round-6 change brought the supplement to thirty-one tables (S1–S31); the table was placed last, so S1–S30 kept their numbers.

A final round-6 pass shortened the manuscript and fixed editorial errors. Repeated conclusions were consolidated to one canonical statement each with cross-references — the Prisma retired-headline story, the version-sensitivity thesis, and the same-SQL "bounds but does not isolate" caveat were each stated once and cross-referenced elsewhere — and the most duplicative auxiliary prose (the open-loop/utilization tail block and the thrice-told multi-worker cluster result) was compressed to its conclusion plus the Supplement table pointer; every trimmed number still appears once in its Supplement table. This cut the body by 232 words, from 11,448 to 11,216, and the total from 14,995 to **14,763** (237 words of headroom under the 15,000 limit). Editorially, the run-in `\{}paragraph` headings were rendering a double period under `elsarticle` (which auto-appends a period to a heading that already ended in one — "the instrument is the contribution.."); the trailing period was removed from all twelve headings so each now renders a single period.

A round-7 methodology-and-scope pass sharpened the claims without changing any measurement: the estimand was renamed from "default-configuration" to "documentation-selected implementation-and-strategy" (choosing the first documented API is not the library's default configuration); the three claim levels — the primary implementation-and-strategy comparison, the same-SQL raw-path bounding control, and the individually non-identified mechanisms — were named and kept separate; "capacity", "saturating throughput", and "overload" were reserved for the deep-fetch sweep, where a throughput knee was actually measured, and the primary matrix's fixed point is described as "high-load" rather than "overload"; and a compact **experiment-scope table** (Supplement Table S32: each experiment's access pattern(s), engine(s), layer count, and repeated runs) was added last, so S1–S31 keep their numbers. The supplement now holds thirty-two tables (S1–S32). A companion CPU-and-statistics pass in the same round demoted the equal-CPU experiment to an exploratory sensitivity check and corrected its "at most a few percent" wording (the ORM per-core values wobble 10–17% run-to-run); added that the bootstrap intervals capture within-campaign variability, not cross-machine generalization; noted that 1–2 ms p99 gaps sit at the millisecond measurement floor; and brought the raw MySQL-insert distributions (Supplement Figure S1) into the body. A final conclusions-and-language pass then limited "access-layer overhead" wording to the combined implementation-and-strategy quantity the design actually measures, replaced the "thin layers are a safer default" recommendation with "benchmark the relation-heavy hot path for the specific application", and added an explicit "configuration-specific" label to the Conclusion (the cautious "in the sources searched" novelty phrasing was already in place). A closing presentation-and-control pass then removed repeated caveats — the standalone same-SQL "bounds not isolates" Discussion paragraph (already stated in Results, the abstract, and the Conclusion), the duplicated horizontal-scaling and "productivity/type-safety not measured" sentences in Practical Guidance, and a redundant single-host restatement — and, in the abstract, dropped the secondary Kendall-tau and matched-utilization results while scoping the open-loop and equal-compute-budget conditions to the deep fetch. A pre-submission consistency sweep confirmed the pinned library versions match `package.json`, the DOI and GitHub link resolve, and every headline number matches its table. That sweep also corrected a CV figure ("within 4.2%", not 4.0%, on the reads), removed the redundant per-cell median CIs from the main-paper paired significance table (Table 7) that had disagreed with the pattern table's bounds, and reconciled two loose prose phrasings ("the faster layers" cluster within 5%; the transactional ordering "broadly tracks" the single-row insert apart from Prisma). These are all rewordings, so the manuscript total was **14,903** words (11,356 body, 97 under the 15,000 limit); the structured abstract is 292 words.

A subsequent editorial pass condensed the main text by roughly **24%** (body 11,356 → 8,669 words; main paper 51 → 43 pages) so it reads as "a reproducible benchmark methodology demonstrated through a dual-engine case study," with the full audit trail in the supplement. Nothing was deleted: the detailed statistical construction (permutation/bootstrap/TOST/ANOVA) became the supplement's *Statistical methods* section, the measurement internals (warm-up, order-invariance, tail estimation, resource sampling) its *Measurement details* section, and the four-pitfall checklist its *Benchmarking-pitfalls checklist* section (main text keeps a named summary); Methodology, Results, Discussion, and Threats were tightened and repeated single-host/version-sensitivity caveats consolidated. The manuscript total is now **12,216** words (8,669 body + 299 abstract + 1,848 references + seven main floats), well under the 15,000 limit; the supplement grew from 26 to 29 pages. All citations were kept in the main text (the supplement has no bibliography), so the reference list is unchanged. This revision is archived as release v1.6.4 (DOI 10.5281/zenodo.21444624).

A round-8 revision subordinated the whole narrative to one primary contribution — a **comparability protocol** for access-layer benchmarking (a semantic-equivalence gate, a documentation-selected estimand bounded by a same-SQL control, and separated operating points) — explicitly distinguished from the reusable artifact and the empirical case study (title, abstract, introduction contribution list, related-work positioning, discussion, conclusion, and highlights). It also addressed the estimand's construct validity (the documentation-selected treatment is a reproducible persona proxy, not a claim about practitioner choice; a same-SQL control that *bounds* rather than *decomposes* the strategy contribution; new evidence in Supplement Table S33), rewrote the same-SQL result in consistently non-causal terms and diagrammed its changed components in a new main-text table (Table 6), and reweighted the RQ2 cross-engine treatment to lead with the concrete rank reversals and the layer$\times$engine interaction magnitudes (on the insert \{}texttt{knex} $1\rightarrow3$ and \{}texttt{drizzle} $2\rightarrow1$ trade the lead while Prisma falls 4$\rightarrow$last; the per-layer PostgreSQL$\div$MySQL advantage scatters 1.6–3.2$\times$ on the insert and 1.4–1.9$\times$ on aggregation; Supplement Tables S27–S28) rather than characterizing the n=7 Spearman coefficients with qualitative labels such as "moderate," which were removed; the coefficients are now reported only as coarse descriptive summaries. These changes added a main-text component diagram (seven $\rightarrow$ eight main floats) and expanded the estimand/RQ2 prose; the body is 9,648 words and the total 13,395, and the structured abstract was tightened (a duplicated version-sensitivity sentence dropped, the reversal detail added) to stay under 300.

A round-8 workload-scope pass then narrowed the one generalizing statement the reviewer flagged — "the ORM penalty concentrates on relation-materializing traffic" — to the tested implementations and patterns: the Discussion now says the documentation-selected ORM spread was largest on the single deep/nested-fetch shape tested and adds that one nested-fetch topology (the fan-out sweep varies its breadth, 0–500 children, but not its depth, schema, or query mix) cannot establish a general distribution of ORM penalties, so the pattern is a benchmark observation to be re-measured on the target workload, not an established law; the Conclusion and the practice paragraph were scoped to "the five tested patterns" / "these probes" to match. The added scoping took the body to 9,739 words and the total to 13,486, still under the 15,000 limit.

A round-8 horizontal-scaling pass then weakened the claim, flagged by the reviewer, that a layer's low per-process throughput is "recoverable by horizontal scaling." The node-cluster check (Supplement Table S24) scales only $1\rightarrow4$ workers on a co-located 16-core host, measures no database-side saturation, and on MySQL the native driver keeps scaling so the gap widens; the one PostgreSQL knee (native `pg`) was previously attributed to becoming "database-bound" with no DB-side evidence. Every claim site (Results, Discussion $\times2$, Threats, the Multi-worker-scaling supplement section, and the S24/S4 table captions in their generators and both copies) now reads "may be addressed by additional application processes until another bottleneck is reached," scoped to the tested $1\rightarrow4$-worker range, with the unsupported "database-bound" attribution removed and the MySQL counter-case and the absence of DB-side saturation measurement stated explicitly. The added caveats took the body to 9,879 words and the total to 13,626, still under the 15,000 limit.

A round-8 language pass then matched the strength of the prose to the replication of the low-run secondary experiments the reviewer flagged (equal-CPU three runs, pool-size three runs, transactional write five runs, and by the same principle the node-cluster and open-loop coordinated-omission checks). Inferential verbs became consistency statements: "an equal-CPU check confirms no layer converts extra cores" -> "is consistent with no layer converting"; the pool-size frontier "shows the ordering preserved" -> "is consistent with the ordering being preserved" and "does not drive the ranking" -> "does not appear to drive the ranking" (in the Threats prose, both supplement pool-size sections, and the S4/S7/S25 captions in their generators and copies); the equal-CPU caption's "is not an artifact of unequal core budgets" -> "does not appear to be an artifact"; the open-loop CO cross-run "confirms the tail ranking holds" -> "is consistent with the tail ranking holding"; the cluster and open-loop "shows" -> "indicates"; and the transactional-write section now states its "median of five runs, exploratory" replication; and the alternative-eager-loading check (median of five) "rules out a mere default-strategy artifact" -> "argues against" and its "deficit is not an artifact of a poor default" -> "does not appear to be an artifact" (Results, Threats, the construct-validity supplement section, and the

S18 caption in its generator and both copies). Deterministic checks (byte-identity, checksums) and the primary-rigor claims (the 25-run matrix, its permutation test, the 25+10-run durability secondary, and the multi-point concurrency sweep's "not an artifact of a single point") keep their stronger wording. The rewording took the body to 9,891 words and the total to 13,638, still under the 15,000 limit.

A round-8 minor-concerns pass addressed five readability/labeling points. The abstract was lightened (MC1/MC4): its Methods clause dropped "equal-compute-budget conditions" (the equal-CPU check is an exploratory secondary), and its Conclusion dropped the retired-five-core-Prisma parenthetical (that result is stated in full in the Discussion and covered by the temporal external-validity threat), so the abstract stays comfortably under the 300-word limit and the core deep-fetch result reads more prominently. "Vendor-independent"/"vendor independence" applied to the study and the coverage axis became "independently authored"/"independent authorship" (MC2), since the manuscript means authorship, not design, independence (the "No (vendor)" prior-art column is unchanged). Drizzle's category label "orm-lightweight" was merged into "orm" (MC3) — an interpretive performance-implying label on an experimentally untested taxonomy — in the Category column of the six pattern/resource tables (both copies), the two generators (`ci-tables.mjs` CAT map and `analyze.mjs` display normalization) and `config.mjs`, with the methodology prose softened to "a query-builder-style ORM"; raw.json's internal category field is left as-is (not reader-facing, so no checksum churn). No secondary material was moved to the supplement (MC5): the manuscript is under the word limit, so density, not length, was the concern, and the R8 caveats are load-bearing for the other points; the abstract trim is the density fix. The body is 9,890 words and the total 13,637, still under the 15,000 limit.

A round-8 novelty-discipline pass held the strongest novelty claim to what a scoping (not systematic) search supports and distinguished the two kinds of novelty. In the Positioning, "none combines [the four properties]" became "we did not identify a study combining [them] — each we found covers at most two or three," followed by an explicit statement that the paper claims only "we did not identify a prior study combining these properties in the documented search," not priority the scoping search cannot establish. In the Introduction's contribution paragraph, the case study is now framed as contributing novel experimental evidence (a dual-engine, throughput-and-p99, documentation-selected comparison not identified in the searched sources) that is configuration-specific, distinct from the novel benchmark infrastructure (the protocol and its harness), which is the more durable contribution. No "first"/priority claim is made anywhere; the audit found only enumerative/temporal "first"s. The introduction's four-axis gap statement was also tidied so the "four coverage axes" are named explicitly (access-layer taxonomy, engine, joint throughput/tail metrics, independent authorship) and the separate HTTP-framework/client-observed gap is presented as "beyond these four axes" rather than as a fourth axis, matching the prior-art table's four columns. The additions took the body to 9,982 words and the total to 13,729, still under the 15,000 limit.

A round-8 closing pass responded to the reviewer's summary and the point-12 list. All six Essential items were already addressed in R8 points 1–7 (sharpen contribution; narrow the five-pattern-workload claims; weaken horizontal scaling; same-SQL as a compound intervention; reduce the weight on the n=7 rank correlations; downgrade low-replicate sensitivity conclusions), and the strongly-recommended distributional write presentation (Supplement Figure S1) and artifact-provenance table (Supplement Table S31) are in place. "Documentation- selected" is kept: the construct-validity treatment already disclaims representativeness, so the term is justified, and a second rename would risk reopening the estimand discussion. For the "over-engineered rhetoric" concern the main text got a light, safe streamlining that removed no load-bearing caveat: the duplicated "productivity/type-safety not measured" qualification in the Practical-guidance paragraph is now stated once, and a few redundant parenthetical "(Section~...)" cross-references in the densest sections (Results, Threats) were dropped where the same section was already cited nearby. No section was reordered and no float moved (the deeper-restructuring option was declined), since the caveats are load-bearing for the Essential points. The trimming took the body to 9,965 words and the total to 13,712, still under the 15,000 limit.

The round-8 revision (all of the above from point 1 through the light streamlining) is archived as release v1.6.5 (DOI 10.5281/zenodo.21455542).

A round-8 follow-up (reviewer point 6.1) stopped calling the same-SQL result a "bound." The reviewer noted that the default-path and raw-path measurements evaluate two different combinations of factors, so — absent interaction assumptions — no ordering theorem makes the raw-path spread a bound on the intrinsic library effect (raw mode may interact differently with each library and may bypass the very mechanisms one wants to characterize). Following the reviewer's strong recommendation, every same-SQL "bound/bounds/bounded" was reworded to a **standardized (same-SQL) contrast / diagnostic** across the abstract, introduction, methodology, results (including the Table 6 caption and the `tab:sameplan` generator + both copies), threats, discussion, conclusion, related work, and supplement; a one-sentence caveat was added in Results ("Because raw mode changes several mechanisms at once and may interact differently with each library, this residual is a diagnostic contrast, not a bound on the intrinsic library effect — no ordering theorem holds without assumptions about interaction effects"). The compound-intervention framing (it changes several mechanisms together; Table 6) is unchanged; only the inequality/ordering implication is removed. The added caveat took the body to 10,084 words and the total to 13,831, still under the 15,000 limit; the structured abstract is 297 words.

The point-6.1 follow-up (same-SQL standardized contrast, not a bound) is archived as release v1.6.6 (DOI 10.5281/zenodo.21456366).

A round-8 follow-up (reviewer point 6.2) strengthened the semantic-equivalence gate rather than renaming it. The reviewer noted that finite-probe byte-equality is not full implementation equivalence and that a non-2xx check catches an HTTP 500 (the MikroORM case) but not a silently incorrect write. A new harness check, `bench/verify-writes.mjs`, validates post-write database state through an independent native-driver connection, for every adapter on both engines: the single-row insert wrote exactly the requested field values and generated identifier (one `posts` row, no stray comments); the transactional write inserted the post and its five comments with the exact fields (one `posts` and five `comments` rows — one intended logical operation); and a transaction whose last comment violates the author foreign key throws and rolls back with no partial state. All adapters pass on both engines, so no throughput number changes. Methodology now describes the write gate and adds an honest finite-probe caveat for reads; the Introduction contribution bullet and the Discussion MikroORM note were updated; `REPRODUCE.md` and the artifact-reproducibility table (Table S31) add the `verify-writes.mjs` command. The added description took the body to 10,264 words and the total to 14,011, still under the 15,000 limit.

The point-6.2 gate strengthening is archived as release v1.6.7 (DOI 10.5281/zenodo.21457569).

A round-8 follow-up (reviewer point 6.3) specified the comparability protocol normatively, abstracted from the case study. A new Methodology subsection, "The comparability protocol" (`sec:protocol`), states it independently of the Express/access-layer setting in five parts: **Inputs** (treatments, workload, output semantics, operating-point definition); **mandatory stages, in order** (correctness oracle → treatment-definition rule → strategy-control design → capacity identification → demand/utilization experiments); **pass/fail** cell-admission (a cell is admitted only if the oracle passes — equivalent non-mutating outputs and a correct post-write state; non-equivalent output, an invalid write, or a degenerate plan disqualifies it); **outputs and their interpretation** (descriptive of the treatment-and-strategy, not a causal decomposition; rankings configuration- and version-specific; only relative within-condition differences travel); and **applicability limits** (byte-equality is a valid oracle only for deterministic, canonically serializable outputs — for unordered collections, floating point, timestamps, nondeterministic identifiers, or differently serialized equivalents the oracle must be a semantic comparator). The Introduction now points to this specification. The added subsection took the body to 10,649 words and the total to 14,396, still under the 15,000 limit.

The point-6.3 protocol specification is archived as release v1.6.8 (DOI 10.5281/zenodo.21457745).

A round-8 follow-up (reviewer point 6.4) narrowed the RQ1 construct and added a performance-conscious co-primary regime. **Part A** reworded the GQM goal from "quantifying its performance cost" to "quantifying the performance of its documentation-selected implementation strategy," tying RQ1 to the selected strategy (the driver-versus-ORM asymmetry — native hand-SQL versus the ORM's documented relation loader — was already stated in Study Design). **Part B** added, on the deep fetch (the only one of the five patterns exposing a documented strategy choice), a second co-primary regime: the performance-conscious regime takes the faster of a layer's two documented loading strategies, admitted only where byte-identical to the documentation-primary output under the `bench/verify.mjs` oracle (a defined, narrow tuning budget that never rewrites SQL, adds caching, or changes the schema). A new harness (`scripts/deepfetch-regimes.mjs`, a warmup pass then 25 timed repeats each) measures both regimes for the four data-mapper ORMs on one footing (`results/deepfetch-regimes.json`, added to the checksum manifest — now thirty-four raw-data files); Study Design defines the tuning budget, Results reports the two regimes, and a new supplement float (Table S34) tabulates them. The finding reinforces rather than overturns the documentation-primary reading: for Sequelize and MikroORM the documented join is already the faster strategy (the two regimes coincide), only Objection on MySQL gains (its documented single-join alternative beats its select-in default), and even there the performance-conscious ORM deep fetch stays below the native driver (a same-harness native reference row, 3,347/2,274 req/s, is included in Table S34). The added prose took the body to 11,008 words and the total to 14,755, still under the 15,000 limit; the structured abstract is unchanged at 297 words. The supplement now holds thirty-four tables (S1–S34) plus two figures.

The point-6.4 follow-up (narrowed RQ1 construct + performance-conscious co-primary regime) is archived as release v1.6.9 (DOI 10.5281/zenodo.21459069).

A follow-up (reviewer point 6.5) demonstrated **capacity identification across all five patterns**, not only the deep fetch. The operating-point protocol requires that capacity be identified so the fixed 50-connection point can be read against it, but capacity sweeps previously existed only for the deep fetch, leaving the relative utilization of the other four patterns unmeasured. A per-pattern concurrency sweep (`bench/scaling-patterns.mjs`, connection ladder 1–200, every layer, both engines → `results/scaling_patterns.json`, the thirty-fifth raw-data file) locates each pattern's throughput knee at or below 50 connections, with median utilization at the 50-connection point of 97–100\{}% across layers (minimum 82\{}%); the fixed point therefore places all five patterns at high utilization of their own capacity, so the cross-pattern p99/spread comparison is one at comparable, measured relative utilization rather than unknown fractions of capacity. Study Design and the RQ3 results now state this, the new Supplement Table S35 reports the per-pattern knee and utilization, and the equal-utilization open-loop tail and exploratory equal-compute check are labeled as remaining demonstrated on the deep fetch. The added prose took the body to 11,156 words and the total to 14,903, still under the 15,000 limit; the structured abstract is unchanged at 297 words. The supplement now holds thirty-five tables (S1–S35) plus two figures.

The point-6.5 follow-up (per-pattern capacity/utilization sweep) is archived as release v1.7.0 (DOI 10.5281/zenodo.21461236).

A follow-up (reviewer point 6.6) corrected an over-strong causal claim without changing any measurement. The Controls subsection previously said four controls "isolate the access layer as the only variable," inconsistent with the treatment being a bundle of differing SQL, query counts, protocols, loading strategies, adapters, and hydration. It now reads that the four controls hold the shared conditions fixed (pool, logical task, physical state, observer) so the *configured access-layer implementation bundle* is the only deliberately varied factor, not the library alone — the bundled differences being the treatment, not confounds. Two related-work sentences using "isolates" for the study's factor were reworded to "treats"/"varies"; the isolation *disclaimers* in the abstract, introduction, and methodology, and the unrelated "write-isolation", "resource-isolation", and "isolate the durability mechanism" uses, were verified correct and kept. The added clause took the body to 11,210 words and the total to 14,957, still under the 15,000 limit (a 43-word margin); the structured abstract is unchanged at 297 words. No supplement float was added, so the supplement stays at thirty-five tables (S1–S35) plus two figures.

The point-6.6 follow-up (corrected the "isolate the access layer" over-claim) is archived as release v1.7.1 (DOI 10.5281/zenodo.21470118).

A follow-up (reviewer **point 7, minor concerns**) applied six wording/labeling corrections, no measurement changed: (1) the version-specific "Prisma 7 sits mid-pack" clause was removed from the abstract (de-loading it and dropping the abstract from 297 to ~292 words); (2) "Express.js remains the most widely used web framework for Node.js" was softened and temporally qualified to "has long been among the most widely used"; (3) Table 1's "Independent?" column was relabelled the more neutral **"Vendor involvement"** (values None/Vendor), the fourth coverage axis renamed to "vendor involvement", with a caption note that it is a source/conflict-of-interest property, not a guarantee that an independently authored benchmark is unbiased; (4) the threats sentence "a hot-cache regime that **maximizes** layer differences" became "expected to **make application/access-layer overhead more visible**"; (5) the primary text now discloses the one Drizzle/MySQL-insert cell measured over 24 rather than 25 runs and that the median/interval accommodate it; (6) "independent runs" was reworded to "repeated runs" in the figure/table captions and REPRODUCE.md (the body already said "repeated runs"), since all repetitions share a host and campaign. To hold the total under the limit, the point-6.6 Controls sentence and the new Table 1 caption note were tightened. The body is 11,250 words and, with the abstract now ~292, the total is **14,990**, under the 15,000-word limit (a 10-word margin; prior accepted revisions ran at 5–9). No supplement float changed; the supplement stays at thirty-five tables (S1–S35).

The point-7 minor-concerns pass is archived as release v1.7.2 (DOI 10.5281/zenodo.21470361).

A follow-up (reviewer **point 8, methodological/statistical assessment** — favourable, "no clear statistical error requiring rejection") applied five precision/scoping fixes, no measurement changed: (1) Methodology now states explicitly that * *"p99" throughout denotes the* median run-level p99* **(the median across replicates of each run's own p99), not the p99 of the pooled request distribution, which the design does not estimate; (2) the within-campaign scope of the bootstrap intervals (already in Study Design) is reinforced where they are used, so they are not read as deployment-general uncertainty; (3) the coarse millisecond p99 resolution is noted not to finely separate near-adjacent layers; (4) following the reviewer's suggestion to simplify the post-hoc rank tests and foreground effect sizes, the paired-significance table (former main-text** Table 7**) was** moved to the supplement (new Table S36)* *with a one-line pointer, and the main text now leads with the paired geometric-mean ratios and per-replicate dominance, noting the tests' inference is conditional on the ranking that selects the pairs; (5) the secondary experiments restricted to layer subsets or few replicates (Supplement Table S32) are labelled* sensitivity evidence *on the tested subset, not claims about all eleven implementations. Moving Table 7 drops the main-text float* *count from eight to seven (1,600 → 1,400 words under the IST rule), which more than offsets the added prose. The body is 11,362 words, the structured abstract ~292, and with seven main-text floats the total is 14,902**, under the 15,000-word limit (a 98-word margin). The supplement gains one table (now thirty-six, S1–S36).

The point-8 statistical-assessment pass is archived as release v1.7.3 (DOI 10.5281/zenodo.21471541).

A follow-up (reviewer **point 9, reproducibility and artifact assessment** — "one of the manuscript's strongest dimensions") verified the five-item pre-submission checklist, all satisfied: the Zenodo tarball byte-matches `git archive` of the tag (identical sha256); every main-text and supplement statistic traces to archived raw data via `MANIFEST.md` (the one disclosed exception, the S2 query-count table, counts SQL statements deterministically, not measurements); `results/environment.txt` plus `experiments/schema/db-config.md` capture kernel/runtime, CPU model/count, NUMA topology, governor, turbo, virtualization (`vmware`), process affinity, lockfile hash, and the database configuration; tables rebuild from raw data with Node built-ins and `npm ci` from the committed lockfile, no network; and the repository carries an explicit dual license (MIT for code, CC BY 4.0 for the paper and data). For the AI-declaration policy point, the single declaration was **split** into the Elsevier-standard *Declaration of generative AI ... in the writing process* (writing only) and a separate *AI-assisted research software* statement (harness implementation/analysis, author responsibility, 19/19 estimator unit tests). The AI declarations are excluded metadata (not article content), and points 1–5 needed no change to the body, so the count is unchanged at body 11,362 / total **14,902**.

The point-9 reproducibility/artifact pass is archived as release v1.7.4 (DOI 10.5281/zenodo.21472130).

A follow-up (reviewer **point 10, novelty and related-work assessment** — the novelty case is "credible but should remain carefully bounded") bounded the *methodological*-novelty claim. The Introduction now states explicitly, after the three preconditions, that "these preconditions are not new to benchmarking in general" — correctness, workload equivalence, warm-up, coordinated-omission correction, capacity curves, and reproducibility are long established in the performance-evaluation literature — and that the contribution is their *domain-specific synthesis and operationalization* for the access-layer genre, not the invention of these principles; the "novel benchmark infrastructure" phrasing was reworded to "the protocol's domain-specific synthesis of established benchmarking principles, and its harness." The scoped "in the sources searched" novelty wording is preserved throughout, as the reviewer recommended, and the same-SQL "bound" is already a "standardized contrast" (point 6.1). The added framing took the body to 11,419 words and the total to **14,959**, still under the 15,000-word limit (a 41-word margin).

The point-10 novelty-framing pass is archived as release v1.7.5 (DOI 10.5281/zenodo.21472308).

A follow-up (reviewer **point 11, presentation** — "generally well written, technically mature, unusually careful about claim scope"; main weakness density) added a **compact formal protocol box** and cut density. Because the protocol is the contribution, the *comparability protocol* subsection of Study Design now opens with a boxed, formal statement (Inputs; the five mandatory stages in order; the cell-admission rule; the output interpretation), which replaces the verbose paragraph-by-paragraph prose and is **inline body text, not a float**, so the exact treatment-selection and admission procedure is in the main text rather than left to `METHODOLOGY.md` and supplementary tables. The most-repeated "protocol, not the ranking, is the contribution" framing was consolidated to one statement per location (the Discussion heading and its closing clause no longer both carry it); the recurring caveats that are load-bearing for earlier points in this review (same-SQL "not a bound", point 6.1; deep-fetch "not a poor-default artifact", point 6.4) are kept once each, since removing them would undo those requested caveats. Net, the box replacement plus the trims took the body to 11,373 words. Under the IST rule the total is body 11,373 + abstract ~292 + references 1,848 + **seven** main-text floats (1,400) = **14,913**, under the 15,000-word limit (an 87-word margin), verified quantitatively as the reviewer asked; the structured abstract satisfies the five-part requirement at ~292 words (< 300).

The point-11 presentation pass (compact protocol box, reduced density) is archived as release v1.8.0 (DOI 10.5281/zenodo.21472649).

A follow-up (reviewer **point 12, the consolidated Essential/strongly-recommended/optional list**) confirmed and closed the list. All five Essential items were already carried out in points 6.1–6.6, 8, and 11 (bound → standardized contrast; write-state oracle; normative protocol; configured-bundle treatment; per-pattern capacity). This pass closed the one remaining gap by separating the protocol box's **mandatory** stages (correctness oracle; treatment-definition rule) from its **recommended** stages (strategy control; capacity identification; demand/utilization), and added a **tie-breaking rule** for "documentation-selected" (where several official paths are equally prominent, as for Drizzle's SQL-style builder versus its relational-query API, select the API at the library's own taxonomy tier and record the rest as alternatives). Following the strongly-recommended density note, the version-sensitivity echoes in the Introduction and Conclusion were trimmed to one brief mention each (the full treatment stays in the Discussion). The optional visual schematic was declined — the compact formal protocol box already carries that content — as were the optional second-host and I/O-bound replications, which stay declared future work. Net of the trims, the mandatory/optional split plus the tie-breaker took the body to 11,431 words; the total is body 11,431 + abstract ~292 + references 1,848 + seven main-text floats (1,400) = **14,971**, under the 15,000-word limit.

The point-12 pass (mandatory/optional protocol split, tie-breaking rule, Essential-list confirmation) is archived as release v1.8.1 (DOI 10.5281/zenodo.21472829).

A follow-up (reviewer **point 15, the final recommendation of* major revision*) **confirmed that the central concern — the same-SQL "bound" pillar — is fully corrected, and reconciled the submission package to it. The only manuscript-body change is one word (Results: the performance-conscious regime now "caps" rather than "bounds" recovery, avoiding any residual "bound" phrasing), so the total is unchanged at** 14,971**. An audit confirmed the same-SQL result carries no positive bound claim (only explicit negations). In the response letter, the stale "bounds" quotation of the Strategy-attribution pillar in the Essential-1 section was corrected to the current "a standardized diagnostic contrast, not an attribution" wording; and the** cover letter was refreshed** to the current title ("A Comparability Protocol for Benchmarking Relational Database Access Layers in Express.js"), leads with the three major-revision changes (same-SQL diagnostic contrast; strengthened write oracle; formalized protocol), and reports the current 14,971-word count and the split AI declaration.

The point-15 pass (major-revision reconciliation: pillar-quote fix, cover-letter refresh) is archived as release v1.8.2 (DOI 10.5281/zenodo.21473103).

A follow-up (reviewer **major concern 6.1** — the protocol's novelty and validation are under-demonstrated) added the two artifacts the reviewer required. A **main-text mapping table (Table 2)** pairs each of the five protocol stages with its generic principle, its access-layer-specific manifestation, the failure if omitted, and the evidence it was load-bearing here — arguing the protocol is complete, individually necessary, distinguishable from generic methodology, and transferable. A **supplement retrospective table (Table S37)** applies the protocol analytically to the eight prior benchmarks in the searched sources, naming for each the absent stages and the one conclusion that becomes uninterpretable (a fairness-audited, non-disputing exercise, built with a multi-agent workflow whose per-benchmark claims were adversarially checked). To stay under the word limit, the peripheral application-tier CPU figure was moved to the supplement (Figure S3) and the Resource-footprint paragraph condensed, so the main-text float count is **un-**

changed at seven and the mapping table is `\{}input` from `tables/` (counted once as a float, its cells not in the body texcount). Both new tables are analytical (authored, not data-derived), noted in `REPRODUCE.md`. The body is 11,428 words and the total **14,968**, under the 15,000-word limit; the supplement now holds thirty-seven tables (S1–S37) plus three figures (S1–S3).

The major-concern-6.1 pass (protocol mapping table + retrospective validation) is archived as release v1.9.0 (DOI 10.5281/zenodo.21473952).

A follow-up (reviewer **major concern 6.2** — "documentation-selected" is a reproducible but weak behavioural construct, and the performance-conscious comparison was relegated to Supplement S34) reframed the construct and moved the comparison into the main text. The general protocol now requires only a *predeclared, reproducible* treatment-selection rule and privileges none (the protocol box's mandatory stage 2, the Introduction's Strategy-attribution contribution, and the Study-Design definition were all genericised, with a new sentence stating that documentation-first is *this case study's* policy). The two co-primary deep-fetch regimes were promoted from Supplement Table S34 to a new **main-text table** (`tab:deepfetch_regimes`) beside the co-primary paragraph in Results. The swap is **net-zero**: the roadmap/outcomes table moved from the main text to the supplement at the vacated S34 slot (new Supplement Table S34, "Results roadmap"), so the main-text float count stays at **seven** and Supplement S35–S37 are unchanged. The reframe added prose that was offset by tightening the co-primary paragraph and the two genericised passages; the body is **11,446** words and the total **14,986**, under the 15,000-word limit. The supplement still holds thirty-seven tables (S1–S37) and three figures; the raw data and its 35-file checksum manifest are unchanged (both moved tables are generated, not raw).

The major-concern-6.2 pass (documentation-selected reframed as a case-study policy; performance-conscious comparison promoted to the main text) is archived as release v1.10.0 (DOI 10.5281/zenodo.21475921).

A follow-up (reviewer **major concern 6.3** — the semantic-equivalence gate is strong but finite and incomplete) sharpened the language and broadened the check. The Study-Design correctness paragraph now distinguishes four levels of the precondition (specification; finite pre-measurement conformance; exhaustive equivalence, infeasible and not claimed; property-based/randomized differential testing) rather than asserting a universal property, and the Introduction's Semantic-equivalence contribution was softened to "byte-identical results on the protocol's conformance suite (fixed probes plus a seeded property-based input sweep)." A new gate (`bench/verify-property.mjs`) draws a fixed-seed sample of 1,000 random post and 1,000 random author IDs plus an explicit edge set and asserts byte-equality to the native-driver oracle on all five read methods: 30,400 comparisons per engine, 60,800 across both, zero divergences (new **Supplement Table S38**, appended after S37 so S1–S37 are unchanged; coverage recorded in `experiments/semantic-equivalence.json`, a DB-derived verification artifact outside the 35-file measurement manifest). The reframe added body prose offset by tightening the correctness and tuning-budget passages; the body is **11,456** words and the total **14,996**, under the 15,000-word limit. The supplement now holds thirty-eight tables (S1–S38) plus three figures.

The major-concern-6.3 pass (four-level semantic-equivalence framing; property-based randomized gate) is archived as release v1.11.0 (DOI 10.5281/zenodo.21479555).

A follow-up (reviewer **strongly-recommended point 6.4** — environmental replication is absent) sharpened the External Validity paragraph without adding a fabricated second environment: the only host available for this revision is the same virtualized machine that produced the primary data, so no same-host re-run is presented as independent-substrate replication. The paragraph now states that the same-host checks bound only *within-host* drift (post-restart re-run up to 16%, ranking intact, Supplement Table S20) and partitions which qualitative conclusions should survive a substrate change (the architectural ones — within-engine ordering and operating-point separation) versus which should not (absolute req/s, gap magnitudes, cross-engine RQ2 ordering), naming an independent-host deep-fetch replication as future work and noting that the protocol's stages are substrate-independent experimental-design controls (Table 2, Supplement Table S37). This is a prose-only change: no data, tables, or floats changed. The added argument was offset by tightening the pool, version-disclosure, multi-worker, and freeze-rule clauses, so the body is **11,453** words and the total **14,993**, under the 15,000-word limit.

The strongly-recommended-6.4 pass (environmental-replication framing; which conclusions survive a substrate change) is archived as release v1.11.1 (DOI 10.5281/zenodo.21479860).

A follow-up (reviewer **strongly-recommended point 6.5** — closed-loop p99 is given more prominence than its interpretation warrants) relabelled the primary p99 and promoted the matched-utilization experiment. The Abstract and Introduction now call it the *high-load response-time p99 under equal closed-loop demand* and immediately distinguish it from the tail at *equal utilization*, stating the coupling at first contact. A new **main-text table** (`tab:tail_regimes`, generated by `scripts/gen-tail-regimes.mjs` from `raw.json` + the utilization sweeps) places the equal-demand p99 beside the matched-50%-utilization p99 per layer. To hold the main-text float count at **seven**, the descriptive five-pattern table moved to the supplement (new **Supplement Table S39**, appended after S38 so S1–S38 are unchanged; net-zero float swap). The new framing was offset by tightening the Operating-point pillar, the RQ1 capacity bullet, the headline-p99 prose, and a duplicated multi-worker sentence; the body is **11,457** words and the total **14,994**, with the structured abstract still ~289 words (< 300). No raw data changed (checksums 35/35); the supplement now holds thirty-nine tables (S1–S39).

The strongly-recommended-6.5 pass (p99 relabelled and distinguished; matched-utilization promoted to a main-text table) is archived as release v1.11.2 (DOI 10.5281/zenodo.21480179).

A follow-up (reviewer **strongly-recommended point 6.6** — the protocol is presented as general but only partially exercised across the workload matrix) defined explicit compliance levels and their coverage. Study Design now names five *compliance levels* (mandatory validity core; capacity characterization; standardized-strategy, utilization-controlled, and resource-normalized extensions) and states that the first two hold across all five patterns on both engines while the three extensions are demonstrated on the deep fetch as a representative-point instantiation, not full-matrix coverage. A new **Supplement Table S40** (`protocol_compliance.tex`, an analytical/authored table appended after S39 so S1–S39 are unchanged) maps each cell to the levels it satisfies. This is prose-plus-one- supplement-table only (no main float added, no raw data changed, checksums 35/35); the ~65 body words were offset by tightening the earlier coverage paragraph, the applicability-limits examples, and duplicated scope-hedging in the Discussion, so the body is **11,454** words and the total **14,991**. The supplement now holds forty tables (S1–S40).

The strongly-recommended-6.6 pass (five explicit protocol compliance levels + a cell-by-level coverage map) is archived as release v1.11.3 (DOI 10.5281/zenodo.21485139).

A follow-up (reviewer **strongly-recommended point 6.7** — some mechanistic language is stronger than the design permits) softened the flagged causal claims to "consistent with" formulations, taking the already-hedged Prisma raw-path conjecture as the model. The two named phrases ("the native driver's lead comes from issuing lean SQL that pushes work onto the database tier" and "how a layer materializes a result graph matters more than how many statements it issues") and the Results "drives the database harder ... so the native driver leads" sentence now read as consistency statements, and "large only when it must materialize" became descriptive. On the reviewer's figure point, both compute-efficiency passages now note that the application-tier CPU figure (Supplement Figure S3) excludes database CPU and point to the combined app-plus-database accounting (Supplement Table S5). A re-audit of the Introduction, Results, Discussion, and Conclusion confirmed the remaining causal statements are either experimentally supported (the durability manipulation with `performance_schema` instrumentation) or already hedged. Prose-only: no data, tables, or floats changed (checksums 35/35). The softening added a few words, offset by trimming duplicated compute-cost prose across Results and Discussion, so the body is **11,449** words and the total **14,986**.

The strongly-recommended-6.7 pass (mechanistic language softened to "consistent with"; app-tier CPU-figure caveat) is archived as release v1.11.4 (DOI 10.5281/zenodo.21485274).

A follow-up addressed two **presentation** remarks. (1) The terminology remark (the alternation among "nine layers," "eleven implementations," "seven portable," and "four native/tuned"): Section 3 (Factors and treatments) now opens with a compact **schematic** — a small tier table (native driver; tuned native baseline; query builder; ORM) with the implementations and engine coverage, followed by the counting arithmetic in one place. The schematic replaces the previous inline prose enumeration and is **non-floating** (a plain centred tabular, not a numbered float), so the main-text float count is unchanged at **seven** and the section does not lengthen. (2) The repetition remark: the fullest redundant restatement of "configuration-specific and version-sensitive" (in the Discussion) was reduced to a back-reference, keeping the caveat in the abstract, Introduction, and Conclusion. Both changes are prose-only (no data, tables, or floats changed; checksums 35/35); the schematic's cells and counting sentence roughly balance the removed enumeration, and the consolidation trims a little, so the body is **11,457** words and the total **14,994**.

The presentation pass (terminology schematic in Section 3; consolidated the perishable-rankings caveat) is archived as release v1.11.5 (DOI 10.5281/zenodo.21485425).

A further presentation pass addressed four caption/prose points. (4) Every throughput/p99 table caption now records the 1 ms p99 resolution and that cells differing by $\{}le 1$ ms are practically unresolved (with a pointer to the paired significance analysis), so the tables no longer invite ranking within-resolution differences. (5) For the bimodal insert cells the main text now reads dispersion from the replicate plot (Supplement Figure S1) and the bootstrap spread rather than the coefficient of variation. (6) After verifying every caption, "within-campaign" was added to the two paired-ratio captions (interaction S28, deep-fetch significance S36) and to the Conclusion-validity sentence, so all 95% bootstrap intervals are labelled within-campaign repeatability, not population-general. (7) The protocol box's Inputs now say "declared workload model (stated access patterns, not a production trace)" rather than "representative workload." The caption changes do not affect the body count (\{}input-ed table captions are counted once as the float, not in the body texcount); the prose additions were offset by trimming a duplicated post-restart clause, so the body is **11,460** words and the total **14,997**. No data changed (checksums 35/35). Note for reproducibility: regenerating the CI tables must respect the analyze engine order (cv/PostgreSQL tables want PostgreSQL-last, significance/resources want MySQL-last), and `gen-r6-tables.mjs` output has hand-tuned captions in the committed tables — so only the intended caption edits were kept; the rest were reverted to HEAD.

The presentation-caption pass (1 ms p99-resolution marking; insert dispersion via replicate plot/bootstrap not CV; within-campaign interval labeling verified; "declared workload model") is archived as release v1.11.6 (DOI 10.5281/zenodo.21486707).

Reviewer **point 8** (follow-up statistical assessment) was addressed across five sub-points, adding two supplement floats (no main float; main-text float count unchanged at seven). (8.1) For the ranking- conditional adjacent-pair tests, the manuscript now leads with a *predefined* contrast set — each portable layer against the native-driver reference (new **Supplement Table**

S41, `gen-native-contrasts.mjs`; paired ratio, within-campaign bootstrap interval, dominance 1.00 on all 25 replicates) — and demotes the adjacent-pair permutation/Wilcoxon tests to a descriptive supplement check. (8.2) The +-5% TOST margin's author-selected/application-dependent/not-preregistered/not-universal caveat, already in the supplement, is now stated at first main-text use. (8.3) Rank correlations remain subordinated to reversals and interaction magnitudes (no change). (8.4) A full HDR recorder is not in the archived data (the load generator records percentiles only), so a new **Supplement Figure S4** (`gen-p99-spread.mjs`) shows the 25 per-run p99 values behind each deep-fetch cell; the main text notes the noisy top-1% estimate and the median-of-run-level estimator, with HDR reconstruction as future work. (8.5) An adversarially-verified inferential-phrasing audit (a multi-agent workflow over all sections) found one genuine over-reach ("confirm" for a general claim), softened to "consistent with." The added main-text prose (predefined-contrast reframe, p99-noise note, TOST caveat) was heavily offset by consolidating the now-duplicated adjacent-pair effect-size recap in Results and Threats; the body is **11,454** words and the total **14,991**. Supplement tables/figures: S1–S41 and S1–S4. No data changed (checksums 35/35).

The point-8 statistical pass (predefined native-driver contrasts S41; p99-spread figure S4; TOST caveat at first use; inferential-phrasing audit) is archived as release v1.11.7 (DOI 10.5281/zenodo.21487096).

Reviewer **point 9** (reproducibility and artifact) required only that the container-engine caveat — the convenience Docker path pins older engines (PostgreSQL 16, MySQL 8.4) and does not reproduce the published numbers, whereas the reference conda path uses PostgreSQL 18.4 / MySQL 9.7.1 — be made conspicuous in the main artifact section, rather than only in the supplement/`REPRODUCE.md`. A bold-led sentence was added to the Data-availability section of both builds (`express_db_access.tex` and `ist_main.tex`). Because the Data-availability section is a **declaration** (excluded metadata, like CRediT/funding/AI declarations), it is **not** part of the counted body, so the manuscript length is unchanged at **14,991** words (body 11,454). No data, tables, or floats changed (checksums 35/35).

The point-9 pass (container-engine reproducibility caveat made conspicuous in the main Data-availability section) is archived as release v1.11.8 (DOI 10.5281/zenodo.21487281).

Reviewer **point 10** (novelty and related work) asked to (a) distinguish methodological novelty from coverage novelty and (b) position the protocol against benchmark-quality/experimental-design frameworks beyond access-layer benchmarking, answering "which existing framework could already generate this protocol?". The Positioning subsection now does both: it labels the four-axis coverage claim as "not the contribution" and adds a paragraph answering the reviewer's question with "none," naming the closest frameworks (Kounev et al.'s systems-benchmarking criteria; Hasselbring's SE-benchmarking standard; Raasveldt and Mühleisen's fair-benchmarking guidance) and showing they operationalize neither the documentation-selected treatment definition nor the semantic-equivalence-gated admission for access layers, and that correctness-before-timing precedents (SPEC/TPC/SQLancer) validate one program or engine against its own reference, not equivalence across competing implementations. The literature analysis was an adversarially-verified multi-agent research workflow (25 agents; 18 candidate frameworks verified, all fall short); a post-freeze search found no new access-layer prior art. **Three references were added** (Kounev et al. 2020; Hasselbring 2021; Rigger and Su 2020, SQLancer), so the reference list grows by ~50 words (to ~1,900); the methodological paragraph (net +~80 body) was offset by tightening the survey subsections (search-terms list, benchmark-methodology, per-study detail across the Colley/Salunke/Lorenz/vendor passages). The body is **11,389** words and the total **~14,985**, under the 15,000 limit. No data, tables, or floats changed (checksums 35/35).

The point-10 pass (methodological positioning vs benchmark-quality frameworks; three references; post-freeze search) is archived as release v1.11.9 (DOI 10.5281/zenodo.21488397).

Reviewer **point 11** (presentation — the manuscript over-defended" and its methodological qualifications dispersed) realized the reviewer's three-layer structure (protocol, then case-study instantiation, then evidence the individual controls matter) through concrete deliverables, without reordering sections. The verbose framed protocol box became a one-page **protocol flow figure** (Figure~1: inputs, the mandatory versus recommended stages, the cell-admission gate, and outputs); a new **estimands table** (Table~3) consolidates, in one place, every estimand and what it does and does not identify; and a new **analysis-roster table** (Table~6) separates the primary analyses from the robustness/sensitivity ones. The repeated defense of the same-SQL comparison was cut to one canonical statement with back-references, and three secondary result tables — the same-SQL components, the deep-fetch regimes, and the matched-utilization tail — were moved to the supplement (Tables S42–S44), catalogued in the new roster. The main-text float count is **unchanged at seven** (six tables plus one figure): the figure and the two new tables offset the three moved out (a net-zero swap). Converting the ~500-word box to a figure and trimming ~200 words of over-defensive prose cut the body from 11,389 to **10,938** words. Under the IST rule the exact word-equivalent total the reviewer asked for is body 10,938 + abstract ~292 + references (65 entries) ~1,900 + seven main-text floats (1,400) = **~14,530 words**, under the 15,000-word limit with a ~470-word margin. No data changed (checksums 35/35); the supplement now holds forty-four tables (S1–S44) plus four figures.

The point-11 presentation pass (protocol flow figure; estimands and analysis-roster consolidation tables; reduced same-SQL defense; three secondary tables moved to the supplement; exact word-equivalent recomputation) is archived as release v1.12.0 (DOI 10.5281/zenodo.21491886).

Reviewer **point 12** (the consolidated Essential / strongly-recommended / optional list) was closed. The six **Essential** items were already in place — E1 domain-specific novelty (point 10), E2 protocol-versus-case-study-policy separation (major concern 6.2), E3 high-load p99 reframe (point 6.5), E4 moderated mechanistic language (points 6.7, 8), E5 the four-level correctness oracle (major concern 6.3), and E6 primary-versus-secondary made unmistakable by the point-11 analysis-roster table — with one reinforcement: for **E3** the Abstract now states explicitly that the near-saturation high-load p99 is chiefly a capacity-and-queueing outcome, not a sub-saturation latency difference. Among the **strongly-recommended** items, **S2** was applied by bringing the two deep-fetch regimes back into the main text as a side-by-side table (**Table 7**), reversing their point-11 move to the supplement since the reviewer wants them in the main paper; the matched-utilization tail (now Supplement Table S43) and the same-SQL components (S42) stay in the supplement. S3 (protocol-to-failure-mode validation) is the existing Table 2 mapping plus the Supplement S37 retrospective; S4 (foreground predefined contrasts and effect sizes) was done in point 8; S5 (shorten materially) was the point-11 density cut. **S1** (independent-environment replication) has no second host available this revision and remains framed as future work. **Optional: O1** a machine-readable protocol checklist (`protocol-checklist.yaml`) now ships in the artifact, encoding the inputs, mandatory and recommended stages, the admission gate, outputs, applicability limits, and compliance levels; **O2** the property-based sweep already reports coverage statistics (60,800 comparisons, zero divergences, Supplement Table S38) and its sample size is a script parameter; **O3** (a cold/I/O-heavier contrasting workload) is declared future work, the manuscript already stating the hypothesis that access-layer spreads compress when database I/O dominates. Returning the regimes table to the main text takes the main-text float count from seven to **eight** (seven tables plus one figure), so the exact IST total is body 10,934 + abstract 299 + references ~1,900 + eight main floats (1,600) = **~14,733 words**, under the 15,000-word limit. No data changed (checksums 35/35); the supplement now holds forty-three tables (S1–S43) plus four figures.

The point-12 pass (Essential list confirmed with the E3 abstract reinforcement; S2 regimes returned to the main text; O1 machine-readable checklist; O2 coverage statistics; S1 and O3 framed as future work) is archived as release v1.12.1 (DOI 10.5281/zenodo.21492694).

Reviewer **point 6.1 (follow-up)** — the most important scientific correction — removed every formulation that reads a mechanistic interpretation into the *compound* same-SQL contrast. The manuscript no longer says "most spread is query strategy, not raw execution" (main-text Table 2, the row the reviewer flagged); it now says most of the documentation-selected spread disappears under the compound raw-SQL standardization, with which component is responsible *not identified*. The causal-sounding pillar/precondition name "strategy attribution" is renamed **"strategy standardization"** throughout (Introduction, Study Design, Conclusion, Figure 1, Table 2, and the machine-readable checklist), the Highlights bullet's "bounds strategy vs machinery" is reworded, and Results now states that attributing the residual to any one component would require a factorial or staged ablation, left to future work. The layer-specific strategy claims (Objection on MySQL) rest on a separate controlled A/B (Supplement Table S18), not the compound contrast, so they stay. Prose-and-caption only: no float count or data changed (checksums 35/35). The ablation caveat and slightly longer non-causal wordings took the body from 10,934 to 10,958 words, so the exact IST total is body 10,958 + abstract 299 + references ~1,900 + eight main floats (1,600) = **~14,757 words**, under the 15,000-word limit.

The point-6.1 follow-up pass (removed the same-SQL mechanistic interpretation; renamed strategy-attribution to strategy-standardization; factorial/ablation noted as future work) is archived as release v1.12.2 (DOI 10.5281/zenodo.21493555). Reviewer **point 6.2** — "intrinsic latency" is not measured. Matched-utilization latency still measures an end-to-end Express request under a specific pool, arrival process, database state, service-time distribution, generated SQL, and hardware; equal fractions of separately-estimated capacity do not isolate an implementation's inherent service-time distribution, so the matched-utilization result does not establish an "intrinsic tail" quantity. Every "intrinsic latency / intrinsic tail / intrinsically worse tail / intrinsic per-request latency" formulation was replaced with a non-overclaiming term — "sub-saturation latency", "latency at a matched offered-load fraction", or "queueing-reduced latency" — across the Abstract (both builds), Introduction, Results, Table 2, the tail-regime and utilization tables, and the paired-p99 significance table. The two supported claims are kept: high-load p99 is heavily affected by proximity to saturation and queueing, and the equal-demand ranking should not be read as a difference in sub-saturation latency. The legitimate "intrinsic library overhead/effect" uses (the RQ1/same-SQL estimand disclaimer, which correctly states the design does not isolate the library's own cost) are unchanged. Prose-and-caption only; no float or data changed (checksums 35/35). The slightly longer wordings took the body from 10,958 to 10,959, so the exact IST total is body 10,959 + abstract 299 + references ~1,900 + eight main floats (1,600) = **~14,758 words**, under the 15,000-word limit.

The point-6.2 pass (replaced "intrinsic latency/tail" with sub-saturation / matched-offered-load terms; kept the legitimate intrinsic-library-overhead disclaimers) is archived as release v1.12.3 (DOI 10.5281/zenodo.21494005).

Highlights (5 bullets, each $\leq$ 85 characters) are in `highlights.tex`.